The Portfolio, Not the Switching Rule: Online Multi-Robot Task Allocation under Failures and Deadline Pressure

Abstract

Online multi-robot task allocation must react to task arrivals, deadlines, and robot failures without excessive reassignment. AHE-MRTA selects among five interpretable allocation paradigms through failure and deadline overrides with a dominance-based fallback; geodesic arrival-time estimates and bounded terminal load repair shape the resulting queues. We evaluate the method in an allocation-only simulator, a stochastic navigation proxy, and a closed-loop ROS 2/Nav2 Gazebo benchmark over multiple fleet sizes. In both simulator settings and at the primary closed-loop scale, AHE-MRTA gives the highest observed completion and deadline performance while satisfying the decision-time budget. Matched ablations show that fixed EDF and orphan-first configurations degrade the primary endpoint, whereas removing the override cascade or fixing the dominant spatial-greedy fallback produces no detectable difference. The evidence therefore supports maintaining access to a suitable allocation paradigm, but not a performance advantage for the implemented switching rule over its dominant fallback.

Index Terms

Multi-robot systems, task allocation, online algorithms, ROS 2, Gazebo.

I. Introduction

Multi-robot systems support concurrent task execution in applications such as infrastructure inspection, warehouse logistics, environmental monitoring, and disaster response. Compared with a single robot, a team can cover a larger area, execute work in parallel, and tolerate individual robot failures [1], [2]. Increasing the fleet size alone, however, does not ensure better mission performance. Tasks must be assigned to capable robots at appropriate times and in an executable order. Multi-robot task allocation (MRTA) provides this coordination between mission objectives, robot capabilities, and execution constraints [3], [4]. In dynamic inspection and waypoint-servicing applications, MRTA decisions directly affect completion, deadline compliance, response time, workload distribution, and recovery after robot loss.

This work addresses a time-extended online MRTA problem. Tasks become active during operation, and each task carries a priority, deadline, and capability requirement. When a robot fails or cannot continue navigation, its unfinished tasks must be reassigned to operational robots. Under the Gerkey–Matarić taxonomy, this is a single-task robot, single-robot task, time-extended (ST–SR–TA) problem [3], [5]. At each allocation event, the system uses the current robot poses, navigation states, and task queues. The decision must ensure that each task has at most one owner, that robot queues remain within capacity, and that robot capabilities satisfy task requirements. Assignments currently being executed by healthy robots must also remain locked. Minimizing the instantaneous robot–task distance is therefore insufficient. The allocator must redistribute orphaned tasks and respond to approaching deadlines. It must also limit computation, unnecessary reassignments, and interruptions to ongoing execution.

Existing MRTA approaches trade off solution scope, decision latency, communication, and interpretability. Optimization and evolutionary methods can search broad assignment spaces, whereas matching, greedy, auction, and consensus methods emphasize different combinations of speed, distribution, and coordination cost [2], [6], [7]. No single family is uniformly appropriate when task urgency, robot availability, and workload change during execution. The comparative implications of these method families are examined in Section II.

Dynamic MRTA research addresses task arrivals, temporal constraints, execution uncertainty, and robot failures [7]–[9]. Mechanism selection has also been studied for MRTA. For example, one approach used spatial scenario features to choose between two auction mechanisms before statically allocated runs [10]. Nevertheless, the measurable incremental benefit of event-level policy selection under changing task and robot states remains insufficiently established. In particular, when post-selection safeguards and navigation–repair components are held constant, the benefit of an online selector over the strongest fixed policy is unclear. If compared methods use different travel estimates, queue rules, or repair stages, reliably separating the selector's effect from the effects of other system components becomes difficult. Allocation quality under idealized motion and mission performance under closed-loop navigation measure different aspects of the system. This study does not assume that online policy switching improves performance in every condition. Instead, it measures the selector's incremental effect under a common execution infrastructure.

For this purpose, we develop AHE-MRTA, a centralized event-triggered selection hyper-heuristic [11]. The method represents online context with four observable variables: task density, the proportion of available robots, deadline pressure, and failure rate. The raw policy choice follows a three-branch priority order. When a robot failure is detected, the allocator activates the policy that prioritizes orphaned tasks. When the failure rule is inactive and deadline pressure exceeds its threshold, the allocator selects earliest-deadline-first. In all other conditions, the choice follows persistent dominance scores updated across allocation events for five policies. A transition constraint is applied after this raw choice. An accepted change sets the transition counter to four. While the counter is positive, a request for a different non-recovery policy is rejected and the counter is decreased by one. The failure-recovery policy is not subject to this constraint and can be activated immediately. This rule prevents small fluctuations in the context variables from causing repeated policy changes. The portfolio contains spatial-greedy, priority-first, earliest-deadline-first, commit-once, and orphan-first policies. All policies use the same queue and ownership state, obstacle-aware geodesic arrival estimates, and in-flight task locks. Before the robot queues are published, a common post-processing stage ensures that each task has one owner, assigns bounded additional work to idle robots, and applies a bounded local load adjustment across queues. The contribution lies in combining established allocation behaviors within a common auditable structure and evaluating the selector's incremental effect through controlled ablations.

The contributions of this paper are as follows:

1) We develop an auditable portfolio architecture that selects among five interpretable allocation policies using four online context variables. The explicit selection rules make the reason for each policy choice traceable.
2) We implement a common allocation infrastructure in which all policies use the same travel estimates, queue state, ownership rules, and final load-balancing repair. This design allows the selector to be evaluated separately from the remaining execution components.
3) We evaluate the method across multiple fleet sizes at three levels: allocation-only simulation, a stochastic navigation proxy, and closed-loop ROS 2/Nav2 Gazebo simulation. This structure separates allocation effects from navigation and closed-loop execution effects.
4) We test selector components and fixed-policy alternatives through matched ablations while holding the other system components constant. This design measures the effect of access to a suitable portfolio policy separately from the effect of the online switching rule.

The remainder of the paper is organized as follows. Section II reviews dynamic MRTA and adaptive allocation. Section III formulates the online problem, and Section IV presents the proposed method. Section V describes the experimental methodology. Section VI reports all simulation, closed-loop, scalability, and ablation results. Section VII interprets the evidence and states its limitations. Section VIII concludes the paper.

II. Related Work

Optimization-based allocation and joint allocation–routing models express constraints explicitly and optimize mission objectives. This broad scope is useful for tightly coupled problems, but solving such models at every online event can become costly as the numbers of tasks and robots increase [2], [12]. Bipartite matching and greedy methods provide deterministic, low-latency decisions. Their behavior is easy to inspect, but a fixed objective may not respond adequately to changing urgency, failures, or workload [3], [8].

Market, auction, and consensus methods distribute coordination through bids and local agreement. They reduce reliance on a central optimizer, but communication, rebidding, and convergence add delay [6], [13], [14]. Evolutionary and self-adaptive methods explore broader assignment spaces, but generally require more computation and may produce run-dependent solutions when stochastic search is used [7], [15]. Thus, these method families balance solution quality, latency, communication, stability, and interpretability differently.

Online MRTA methods address dynamic task arrivals, temporal constraints, execution uncertainty, robot failures, and changing communication through matching, adaptive grouping, or self-adaptation [7]–[9], [16]. Most retain one allocation mechanism and update its parameters or repeat its decisions as the state changes. Comparisons between adaptive and fixed methods may therefore also change the travel model, queue discipline, or post-assignment repair.

Mechanism selection offers a complementary approach. Schneider et al. used spatial features to choose between two auction mechanisms before static exploration runs [10]. In contrast, this study examines event-level selection as deadline and failure conditions change. This study does not claim to introduce the first MRTA portfolio or a new solver for each policy. Its contribution is an auditable online pipeline and a matched comparison of selector components and fixed policies under common safeguards, travel estimates, and repair procedures.

III. Problem Formulation

A. System Model and Constraints

Let $\mathcal{R} = \{r_1, \ldots, r_n\}$ be the robot set. Task τ is

$$\xi_\tau = (p_\tau, t^{\mathrm{a}}_\tau, d_\tau, \pi_\tau, c_\tau),$$

where p_τ, t^{a}_τ, d_τ, π_τ, and c_τ are its position, activation time, deadline, priority, and capability requirement. At event t_k, $\mathcal{T}_k$ is the set of active, unfinished tasks. Each robot has a pose, health state, navigation state, and ordered task queue. The available set $\mathcal{R}^{\mathrm{av}}_k$ contains healthy robots that are not navigation-stuck and have remaining queue capacity. In-flight pairs $\mathcal{I}_k$ are locked, so replanning changes only future queue entries.

TABLE I
FIVE HORMONES, FIVE ALLOCATION PARADIGMS.

Hormone	Paradigm	Inner mechanism
H_SPATIAL	spatial_greedy	nearest-feasible + reject
H_CRIT	priority_first	priority-tiered LSA
H_TEMP	edf_strict	EDF bipartite (multi-phase)
H_STAB	commit_once	hard sticky, no reassign
H_RECOV	orphan_first	orphan rescue + bipartite

The binary variable $x^k_{r\tau} = 1$ indicates that task τ is assigned to robot r after event t_k. The feasible set is

$$\begin{aligned}
\mathcal{X}_k = \Big\{x \in \{0,1\}^{n\times|\mathcal{T}_k|} : \; & \sum_{r\in\mathcal{R}} x^k_{r\tau} \le 1, && \forall \tau \in \mathcal{T}_k, && (1)\\
& \sum_{\tau\in\mathcal{T}_k} x^k_{r\tau} \le Q, && \forall r \in \mathcal{R}, && (2)\\
& x^k_{r\tau} = 1 \Rightarrow \big(c_\tau \subseteq \mathrm{cap}(r)\big), && \forall r, \tau, && (3)\\
& x^k_{r\tau} = 0, && \forall r \notin \mathcal{R}^{\mathrm{av}}_k,\ \forall \tau, && (4)\\
& x^k_{r\tau} = x^{k-1}_{r\tau}, && \forall (r,\tau) \in \mathcal{I}_k \Big\}. && (5)
\end{aligned}$$

These constraints enforce single ownership, queue capacity, capability compatibility, and in-flight locks. Failed or stuck robots receive no new tasks. Their unfinished tasks become eligible for reassignment.

B. Event-Level Allocation Objective

For candidate pair (r, τ), predicted arrival $\hat{a}^k_{r\tau}$ combines the current queue endpoint, queue delay, and cached geodesic travel time. Unreachable pairs are infeasible. A pair involving a task with a finite deadline is also infeasible when

$$\hat{a}^k_{r\tau} > d_\tau + s^k_{r\tau} \quad \Longrightarrow \quad (r,\tau) \notin \mathcal{F}_k, \tag{6}$$

where $\mathcal{F}_k$ is the eligible set. The slack $s^k_{r\tau}$ is zero for new tasks and may be positive only for bounded rescue cases.

The method excludes the trivial all-zero plan by first maximizing the number of assigned tasks:

$$\mathcal{X}^\star_k = \arg\max_{x\in\mathcal{X}_k} \sum_{r\in\mathcal{R}} \sum_{\tau\in\mathcal{T}_k} x^k_{r\tau}. \tag{7}$$

It then minimizes the cost defined by the selected policy:

$$x^k \in \arg\min_{x\in\mathcal{X}^\star_k} \sum_{r\in\mathcal{R}} \sum_{\tau\in\mathcal{T}_k} C^k_{r\tau} x^k_{r\tau}, \qquad C^k_{r\tau} = \infty \text{ if } (r,\tau) \notin \mathcal{F}_k. \tag{8}$$

Repeated linear-sum-assignment (LSA) passes approximate this event-level objective, and unassigned tasks remain open. The formulation does not optimize the complete mission offline.

IV. PROPOSED AHE-MRTA METHOD

A. Architecture and Policy Portfolio

AHE-MRTA is a centralized selection hyper-heuristic over five fixed allocation policies [11]. At each allocation event, it reads the latest context and applies failure and deadline overrides. If neither override applies, a five-dimensional dominance state determines the policy. A dwell constraint limits frequent policy changes. The selected policy uses shared queue, ownership, and geodesic-travel data. Common safeguards and bounded load repair are applied before robot-specific queues are published (Fig. 1). Persistent state comprises D_k, the active policy, the dwell counter, in-flight locks, previous ownership, and orphaned tasks.

The five policies share queue, ownership, and geodesic-travel infrastructure but differ in task ordering, eligibility, cost shaping, and incumbent treatment. They approximate Eq. (8) through greedy assignment or repeated masked LSA passes. None optimizes the complete mission globally.

Spatial-greedy selects the feasible task with the nearest predicted arrival. Priority-first applies LSA within priority tiers. EDF-strict orders tasks by deadline, commit-once retains healthy incumbents, and orphan-first reallocates failed or aborted work before new tasks. These policies follow established metric, tiered, temporal, and resilient MRTA patterns [1], [13], [17].

B. Context Modeling and Policy Selection

The method represents the latest observable state as

$$c_k = \left[c_{1,k}, c_{2,k}, c_{3,k}, c_{4,k}\right]^\top \in [0,1]^4, \tag{9}$$

where $n = |\mathcal{R}|$, $m_k = |\mathcal{T}(t_k)|$, and $\mathcal{R}_k^{\text{alive}}$ contains robots that are neither failed nor stuck:

$$c_{1,k} = \min\left(1, \frac{m_k}{\max(1,n)}\right), \qquad \text{task density}, \tag{10}$$

$$c_{2,k} = \frac{|\mathcal{R}_k^{\text{alive}}|}{\max(1,n)}, \qquad \text{robot availability}, \tag{11}$$

$$c_{3,k} = \frac{|\{\tau \in \mathcal{T}(t_k) : d_\tau - t_k \le \Delta_d\}|}{\max(1,m_k)}, \qquad \text{deadline pressure}, \tag{12}$$

$$c_{4,k} = \frac{|\mathcal{R}_k^{\text{f}}|}{\max(1,n)}, \qquad \text{failure rate} \tag{13}$$

Here $\Delta_d = 60$ s, and $c_{3,k}$ excludes tasks without finite deadlines. The context manager updates this vector every 2 s. An allocation event uses the latest completed update rather than waiting for a new one.

The preliminary policy choice $\tilde{p}_k$ is

$$\tilde{p}_k = \begin{cases} \text{H\_RECOV}, & c_{4,k} > 0.05, \\ \text{H\_TEMP}, & c_{4,k} \le 0.05 \ \wedge \ c_{3,k} > 0.50, \\ \arg\max_i D_{i,k+1}, & \text{otherwise.} \end{cases} \tag{14}$$

Recovery takes precedence when both thresholds are exceeded. H_TEMP is used if D_k is unavailable or does not distinguish between policies. After an accepted change, the next $K_{\text{dwell}} = 4$ requests for another non-recovery policy are deferred. H_RECOV can be activated immediately.

Each policy has a fixed context prototype $V_i \in [0,1]^4$ with cosine compatibility

$$v_{i,k} = \frac{V_i^\top c_k}{\|V_i\|_2 \, \|c_k\|_2} \quad \text{and} \quad \mathbf{v}_k = [v_{1,k}, \dots, v_{5,k}]^\top, \tag{15}$$

where a zero-norm vector has zero compatibility. The released configuration provides all prototype values.

When neither override applies, the simplex-normalized state D_k is updated as

$$D_{k+1} = \text{Proj}_{\Delta^4}\Big(\text{clip}_{[0,1]}\left[\alpha D_k + \eta A D_k - \lambda S D_k + \beta \mathbf{p}_k + \gamma \mathbf{v}_k + \delta \mathbf{b}_k\right]\Big) \tag{16}$$

where A and S encode sparse support and suppression, $\mathbf{p}_k$ contains short-horizon completion feedback, and $\mathbf{b}_k$ supports failure recovery [18]. Projection applies non-negative ℓ_1 normalization with a uniform fallback. This fixed update rule is not learned and has no optimality guarantee.

The dominance state also determines weights for travel time, priority, battery, load, failure, deadline urgency, and recovery through a fixed policy map M:

$$w_k^{\text{eco}} = \text{softmax}\left(\frac{M D_{k+1}}{T_w}\right), \qquad w_k = 0.3 w_0 + 0.7 w_k^{\text{eco}}, \tag{17}$$

where $T_w = 0.3$. During recovery, the weights move toward w_{rec} by $\theta_k = \min(0.80, 0.50 + 0.60 c_{4,k})$. Deadline pressure multiplies the urgency channel by 1.5. The battery feature is zero in the evaluated system.

C. Shared Allocation and Execution Safeguards

All policies use the same normalized arrival, priority, battery, load, failure, deadline, and recovery features in the assignment cost:

$$\begin{aligned} C_{r\tau}^k &= \rho(\pi_\tau)\big[w_d A_{r\tau}^k + w_p P_\tau + w_b B_r^k + w_l L_r^k \\ &\quad + w_f F_r^k + w_t U_\tau^k + w_r R_r^k + \Psi_{r\tau}^k\big], \\ \rho(\pi) &= 1 - 0.1(\pi - 1). \end{aligned} \tag{18}$$

Here, $A_{r\tau}^k = \min(1, \hat{a}_{r\tau}^k/220)$, and $\Psi_{r\tau}^k$ contains deadline, capability, incumbent, reassignment, and capacity terms. The selector supplies w, whereas the active policy determines task order and eligibility. Section VII-B discusses known feature-saturation and priority-scaling artifacts.

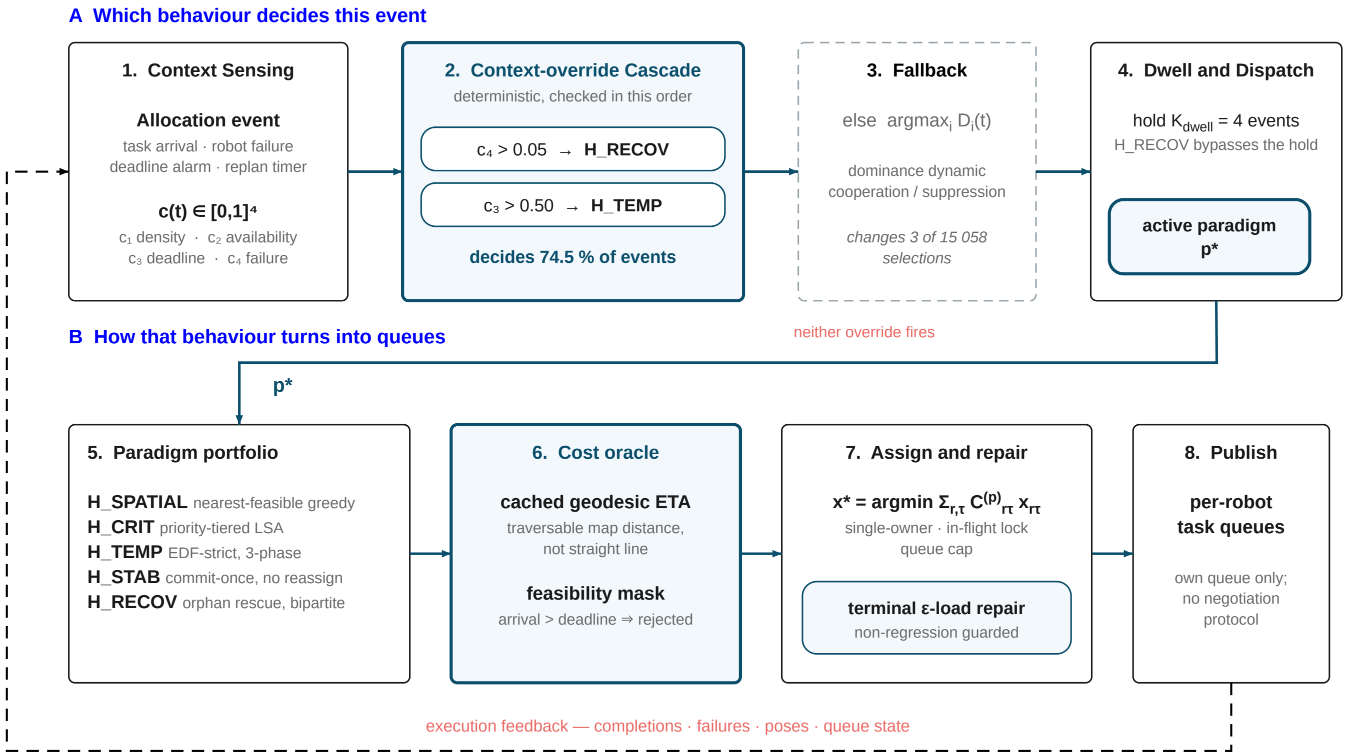

Fig. 1. AHE-MRTA event cycle. Context overrides and the dominance fallback select a policy subject to the dwell constraint. Policy-specific assignment uses shared geodesic travel and queue state. Publication safeguards and bounded terminal repair then produce the robot queues.

All policies preserve single ownership and in-flight locks. At-risk tasks receive a nonlinear urgency term. An assignment held by a healthy robot changes only if the predicted arrival improves by at least 10%. Orphaned, rescue, and urgent tasks are exempt from this threshold.

After policy assignment, a bounded backfill orders pending tasks by priority and deadline and offers them to idle, healthy robots by nearest distance. It may bypass deadline admission only when a task and a robot would otherwise both remain idle. It never preempts an executing task.

Travel estimates use a cached eight-connected geodesic map of the occupancy mask inflated by 0.55 m. The map provides distance d_G, and unreachable pairs are infeasible. For queue $Q_r = (q_1, \ldots, q_j)$, $e_r(Q_r)$ is the robot pose when the queue is empty and the final queued goal otherwise. The estimated time of arrival is

$$\hat{a}^k_{r\tau} = t_k + \frac{d_G(e_r(Q_r), p_\tau)}{v_r} + j\,(h + s_\tau), \tag{19}$$

where s_τ is the service time, $h = 22$ s, and the effective speed is $v_r = 0.22$ m/s. Admission, assignment cost, and repair use the same estimate.

The terminal repair considers only local moves of newly assigned tasks. Given predicted deadline misses M, distance D_G, Jain balance J, robot load B_r, and remaining completion time F, an admissible candidate satisfies

$$\begin{aligned} \mathcal{N}(x^0) = \big\{x : M(x) \le M(x^0), \quad & D_G(x) \le (1+\epsilon) D_G(x^0), \\ J(x) \ge J(x^0), \quad & \max_r B_r(x) \le \max_r B_r(x^0), \\ F(x) \le F(x^0)\big\}, \quad & \epsilon = 0.10. \end{aligned} \tag{20}$$

Candidates are ranked by balance, maximum load, load range, and distance. The repair runs only when the number of active tasks is at most three times the number of healthy robots, and it never cancels an executing goal. Travel estimation and repair affect the assignment and its execution cost but not the selector state.

D. Algorithm and Computational Complexity

Algorithm 1 summarizes one allocation cycle. Only robot queues are published.

Algorithm 1 AHE-MRTA Event Cycle

Require: $\mathcal{R}$, $\mathcal{T}(t_k)$, previous queues, D_k, p_{k-1}, dwell counter h_k; fixed A, S, V, M
$c_k \leftarrow \text{LatestPublishedContext}()$
Update the feasibility mask, orphan pool, and in-flight locks
$\mathbf{v}_k \leftarrow (\cos(V_i, c_k))_{i=1}^{5}$; $\mathbf{p}_k, \mathbf{b}_k \leftarrow \text{PerformanceAndFailureSupport}()$
$D_{k+1} \leftarrow \text{UpdateDominance}(D_k, \mathbf{v}_k, \mathbf{p}_k, \mathbf{b}_k, A, S)$ {Eq. (16)}
$w_k \leftarrow \text{GenerateWeights}(D_{k+1}, c_k, M)$ {Eq. (17)}
$\tilde{p}_k \leftarrow \text{RawChoice}(c_k, D_{k+1})$ {Eq. (14)}
$(p_k, h_{k+1}) \leftarrow \text{ApplyDwell}(\tilde{p}_k, p_{k-1}, h_k)$
$x_k \leftarrow \text{Paradigm}_{p_k}(\mathcal{R}, \mathcal{T}(t_k), w_k)$ {Table I}
$x_k \leftarrow \text{AntiIdleBackfill}(x_k)$
if in the terminal repair phase **then**
  $x_k \leftarrow \text{EpsilonLoadRepair}(x_k, d_G, \epsilon)$
end if
$x_k \leftarrow \text{SingleOwnerAndInFlightLock}(x_k)$
Publish robot-specific queues and store (D_{k+1}, p_k, h_{k+1})

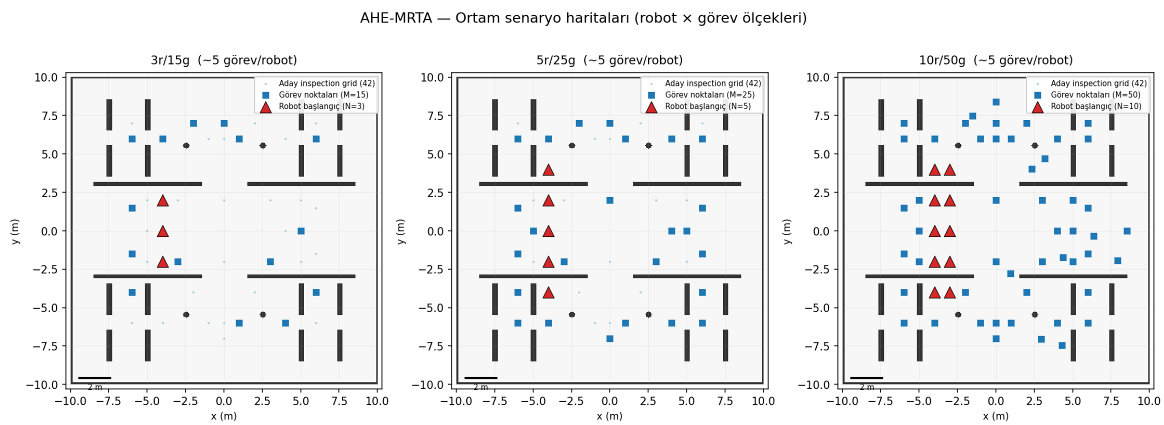

Fig. 2. Arena configurations by scale. **Left:** 3r/15t with robots in one central column. **Center:** 5r/25t with two spawn columns. **Right:** 10r/50t with all five columns active. Blue stars denote task goals for seed 1, and colored triangles denote robot spawns. All waypoints are obstacle-free.

All selector, prototype, matrix, and cost parameters are fixed across scenarios and scales. They are not relearned between experiments. The supplementary configuration provides their complete values.

Selector updates require $O(1)$ time. One square LSA pass requires $O(\max(n, m_k)^3)$ time. Geodesic queries are cached, and repair uses a bounded number of local passes.

V. Experimental Methodology

A. Evaluation Questions and Design

The experiments address three research questions. RQ1 evaluates whether the complete pipeline improves allocation quality under robot failures and deadline pressure. RQ2 examines whether this behavior persists after adding navigation uncertainty and closed-loop execution. RQ3 separates the effects of the policy portfolio, online selector, travel estimate, and repair components.

Three evaluation settings separate these effects. Deterministic geodesic execution isolates allocation and queue order. A stochastic, position-dependent navigation proxy adds goal failures and retries. Closed-loop Gazebo/ROS 2 executes physics and Nav2 end to end. The results are reported separately and interpreted jointly in Section VII.

B. Experimental Platform and Test Conditions

The implementation uses Gazebo Harmonic, ROS 2 Jazzy, and Nav2 [19]–[21]. Each TurtleBot3 Waffle Pi [22] runs an independent Nav2 stack. Ground-truth-anchored localization isolates allocation from localization error. In the result tables, AHE-MRTA* denotes the configuration with geodesic arrival-time estimation and terminal repair enabled. The supplementary artifact provides the hardware, launch, lifecycle, namespace, and logging details.

The 20×20 m inspection arena contains divider walls, pillars, cylindrical obstacles, and 42 admissible waypoints. Each seed deterministically selects task goals from these waypoints (Fig. 2).

TABLE II
GAZEBO EVALUATION SCALES

Scale	Robots	Tasks	Density	Total exps
3r-low	3	9	3.0 t/r	60
3r-mid	3	15	5.0 t/r	60
3r-high	3	24	8.0 t/r	60
5r-mid	5	25	5.0 t/r	240
10r-mid	10	50	5.0 t/r	60
Total Gazebo experiments				**480**

The experiments use 3-, 5-, and 10-robot fleets (Table II). The primary 5r/25t cells contain 20 seeds for each method and scenario. The other cells contain five seeds.

All fleet sizes use the same map and Nav2 parameters. The artifact provides launch-level lifecycle details.

The nominal mission horizon is $T_{\max} = 900$ s. Each task draws priority $\pi_\tau \sim \mathcal{U}\{1, 2, 3\}$, service time $\mathcal{U}[2, 8]$ s, and deadline $d_\tau = t_0 + \Delta_\tau$.

- **robot_failure:** all tasks arrive at t_0 with $\Delta_\tau \sim \mathcal{U}[90, 300]$ s. One designated robot fails at 45 ± 5 s.
- **mixed_stress:** the same failure is combined with arrivals at 0, 30, and 60 s and $\Delta_\tau \sim \mathcal{U}[45, 150]$ s.
- **deadline_pressure:** all tasks arrive at t_0, no robot fails, and $\Delta_\tau \sim \mathcal{U}[36, 120]$ s.

Task attributes and failure times are controlled by the seed and shared across methods. The failed robot index is fixed.

C. Baseline and Ablation Configurations

The benchmark compares operational adaptations of three MRTA families within the same event and Nav2 framework. Queue capacity is policy-specific. AHE uses $Q = 2$ with one additional slot during repair. BiG-MRTA and Consensus-DBTA use $Q = 5$, whereas RoSTAM-EA has no within-event bound. The method names refer to benchmark implementations rather than complete independent reproductions.

BiG-MRTA [8] uses maximum-weight bipartite matching as a deterministic, low-latency method with stable ownership. RoSTAM-EA [7] re-optimizes candidate assignment vectors at each event through stochastic evolutionary search. Consensus-DBTA [14] uses two-best-task bidding and simulated maximum consensus to represent distributed allocation with limited rebidding.

Because the baselines differ in allocation logic and queue depth, the four-method benchmark compares complete configured pipelines. The controlled proxy and closed-loop ablations in Section VI-C provide component-level attribution.

The proxy ablation replaces the selector with each fixed portfolio policy. It also disables the context overrides and recovery boost separately while holding all other components fixed. The matched closed-loop ablation compares the full method, no-override variant, fixed EDF, and fixed orphan-first using common seeds. Fixed spatial-greedy is reported as an exploratory configuration because it was added after the planned four-configuration design.

D. Metrics and Statistical Analysis

The sets of completed tasks $\mathcal{T}^{\text{done}}$ and requested tasks $\mathcal{T}^{\text{req}}$ define the completion rate (CR) and deadline-violation rate (DVR):

$$\max\ \text{CR} = \frac{|\mathcal{T}^{\text{done}}|}{|\mathcal{T}^{\text{req}}|}, \qquad \min\ \text{DVR} = \frac{|\{\tau : d_\tau < \infty\ \wedge (t^{\text{c}}_\tau > d_\tau\ \vee\ \tau \notin \mathcal{T}^{\text{done}})\}|}{\max\big(1, |\{\tau : d_\tau < \infty\}|\big)}. \tag{21}$$

CR and DVR are the primary outcomes. We also report the preregistered composite score $\text{CR}(1 - \text{DVR})$ and the secondary outcomes of delay, makespan, recovery, Jain balance, churn, latency, payload, and distance. DVR counts both late and unfinished tasks. Delay for unfinished tasks is censored at $T_{\max}$. The composite score rewards both completion and punctuality and is zero if either component is zero. The AHE decision-time target is 50 ms per allocation event.

Pairwise two-sided Mann–Whitney U tests use Bonferroni correction within each scenario for three baselines and seven metrics. Cliff's δ measures effect size. Primary 5r cells have $n = 20$, pooled 3r cells have $n = 15$, and 10r cells have $n = 5$. Allocation-only analyses use 100 seeds, and proxy analyses use 500 seeds. Both are analyzed separately from the Gazebo results.

VI. RESULTS

A. Allocation-Only and Navigation-Proxy Results

Under deterministic geodesic execution, AHE-MRTA completes all tasks in each of the nine scale–scenario cells, with 100 seeds per cell. The priority-weighted on-time fitness distinguishes the methods because CR is uniform. At the primary 5r/25t

TABLE III

ALLOCATION FITNESS (PRIORITY-WEIGHTED ON-TIME COMPLETION), 500 SEEDS, 5 ROBOTS / 25 TASKS. **LEFT:** PERFECT-NAVIGATION PLANE, WHICH ISOLATES ALLOCATION QUALITY. **RIGHT:** THE STOCHASTIC NAVIGATION PROXY, WHERE NAV2 GOAL FAILURE CAN ALSO COST A TASK. AHE-MRTA LEADS EVERY SCENARIO ON BOTH PLANES; EVERY PAIRED-WILCOXON COMPARISON AGAINST A BASELINE IS SIGNIFICANT EXCEPT ONE (ALL $p < 0.143$ EXCEPT AHE–BIG UNDER ROBOT FAILURE, WHICH IS A TIE). **BOLD** MARKS THE BEST PER COLUMN.

	Perfect nav (alloc. quality)			Stochastic nav (resil.)		
Method	RF	MS	DP	RF	MS	DP
AHE-MRTA*	**0.994**	**0.727**	**0.840**	**0.384**	**0.263**	**0.349**
BiG-MRTA	0.980	0.668	0.733	0.380	0.229	0.284
RoSTAM-EA	0.935	0.643	0.679	0.333	0.225	0.227
Cons-DBTA	0.978	0.642	0.729	0.364	0.217	0.258

TABLE IV

MAIN COMPARISON AT THE PRIMARY 5-ROBOT / 25-TASK SCALE; n=20 RUNS (SEEDS) PER CELL. BEST VALUE PER COLUMN (WITHIN SCENARIO) IN **BOLD**. SIGNIFICANCE VS AHE-MRTA*: $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$ (MANN–WHITNEY U, BONFERRONI-CORRECTED WITHIN EACH SCENARIO FAMILY OF 21 TESTS).

Method	**CR↑**	**Delay↓(s)**	**RecT↓(s)**	**Preempt↓**	**Churn↓**
Scenario: robot_failure					
AHE-MRTA*	**1.000 ± 0.000**	99.7 ± 7.6	83.7 ± 43.8	0.00 ± 0.00	0.006 ± 0.015
BiG-MRTA	0.956^{***} ± 0.036	104.4 ± 30.1	**73.1 ± 33.9**	0.00 ± 0.00	**0.004 ± 0.013**
RoSTAM-EA	**1.000 ± 0.000**	92.6 ± 10.8	74.5 ± 35.8	0.00 ± 0.00	0.012 ± 0.019
Cons-DBTA	**1.000 ± 0.000**	**86.0^{**} ± 11.2**	117.9 ± 29.4	0.00 ± 0.00	0.006 ± 0.015
Scenario: mixed_stress					
AHE-MRTA*	**1.000 ± 0.000**	65.3 ± 9.1	41.9 ± 17.6	0.00 ± 0.00	0.014 ± 0.020
BiG-MRTA	0.656^{***} ± 0.060	311.2^{***} ± 46.1	**22.8 ± 23.7**	0.00 ± 0.00	0.018 ± 0.029
RoSTAM-EA	**1.000 ± 0.000**	**61.9 ± 10.1**	85.3 ± 56.0	0.00 ± 0.00	**0.002 ± 0.009**
Cons-DBTA	0.998 ± 0.009	72.8 ± 16.4	81.5 ± 82.3	0.00 ± 0.00	0.032 ± 0.098

scale, AHE-MRTA obtains 0.994, 0.727, and 0.840 under robot failure, mixed stress, and deadline pressure. The strongest baseline obtains 0.980, 0.668, and 0.733, respectively (Table III).

The stochastic navigation proxy adds position-dependent goal failures and retries. Across 500 seeds, AHE-MRTA has the highest mean in all three scenarios. Eight of the nine paired baseline comparisons are statistically significant. The exception is BiG-MRTA under robot failure (+0.004, $p = 0.14$).

The AHE-MRTA proxy fitness values are 0.384, 0.263, and 0.349, compared with 0.380, 0.229, and 0.284 for the strongest baseline.

Across the 3-, 5-, and 10-robot fleets, with 100 proxy seeds per cell, the mean margins over the best baseline are 3.3, 3.6, and 2.7 percentage points. The margin therefore does not increase with fleet size. Decision latency at 10 robots remains approximately 0.54 ms. The released artifact includes the complete distributions for each fleet size.

B. Closed-Loop Gazebo Evaluation

The closed-loop benchmark contains 480 runs: 180 with 3 robots, 240 at the primary 5r/25t scale, and 60 with 10 robots. Tables IV and V report the primary cells.

Under deadline pressure with 5 robots, AHE-MRTA* attains a composite score of $\mathrm{CR}(1 - \mathrm{DVR}) = 0.650$, compared with 0.578 for the strongest baseline. All three reactive methods complete almost every task. Their difference is therefore deadline compliance: DVR is 0.350 for AHE-MRTA and 0.422 for Consensus-DBTA.

At the primary scale, AHE-MRTA completes all tasks in every scenario. Its composite score is 0.986 under robot failure and 0.616 under mixed stress.

These gains involve measurable costs. Under robot failure, AHE-MRTA has greater delay than Consensus-DBTA (99.7 versus 86.0 s, corrected $p = 0.002$, $\delta = 0.72$). Its decision latency of 0.81–1.33 ms is also higher than that of commit-once BiG-MRTA. It remains below both the 50-ms target and the latency of RoSTAM-EA. Redispatch is at most 0.014 per task, and no executing task is preempted.

At the primary 5-robot scale ($n = 20$ per cell), 20 of 63 comparisons remain significant after correction. Fifteen favor AHE-MRTA, and five favor a baseline. The artifact provides detailed effect-size, efficiency, trajectory, and timeline analyses.

C. Ablation Analysis

The first proxy ablation evaluates the selector and fixed policies defined in Section V-C. It uses 100 seeds at 5r/25t and holds the shared allocation infrastructure fixed.

TABLE V

DEADLINE SCENARIO RESULTS (DEADLINE_PRESSURE) AT THE PRIMARY 5-ROBOT / 25-TASK SCALE; n=20 RUNS (SEEDS) PER CELL. DVR: DEADLINE VIOLATION RATE. BEST VALUE PER COLUMN IN **BOLD**. SIGNIFICANCE VS AHE-MRTA* (BONFERRONI-CORRECTED MANN-WHITNEY U).

Method	CR↑	Delay↓(s)	DVR↓	Preempt↓
AHE-MRTA*	**1.000 ± 0.000**	78.5 ± 7.7	**0.350 ± 0.090**	0.00 ± 0.00
BiG-MRTA	0.664^{***} ± 0.059	327.0^{***} ± 46.5	0.402 ± 0.073	0.00 ± 0.00
RoSTAM-EA	**1.000 ± 0.000**	77.8 ± 7.3	0.498^{***} ± 0.073	0.00 ± 0.00
Cons-DBTA	**1.000 ± 0.000**	**72.6 ± 9.3**	0.422 ± 0.107	0.00 ± 0.00

TABLE VI

SELECTOR ABLATION (STOCHASTIC NAVIGATION PROXY, GEODESIC EXECUTION ORACLE, 5R/25T, 100 SEEDS). ON THE CORRECTED SCENARIO PARAMETERS THE VARIANTS SPAN 0.101 IN MEAN FITNESS. REMOVING THE CONTEXT OVERRIDES LEAVES THE SELECTOR ESSENTIALLY UNCHANGED, AND FIXED SPATIAL-GREEDY IS AHEAD OF IT ON THIS PLANE—THE CLOSED-LOOP ABLATION (SECTION VI-C) TESTS BOTH OBSERVATIONS.

Variant	Rob. fail.	Deadline	Mix. stress	Mean
Full EDPS	0.383	0.348	0.255	0.329
fixed: spatial	0.421	0.358	0.265	**0.348**
fixed: priority	0.357	0.298	0.274	0.310
fixed: edf	0.333	0.296	0.276	0.302
fixed: commit	0.344	0.186	0.210	0.247
fixed: orphan	0.318	0.259	0.254	0.277
no-override	0.384	0.331	0.254	0.323
no-recovery-boost	0.382	0.348	0.255	0.328

Full selector mean 0.329; best variant 0.348.

Removing the overrides or recovery boost changes proxy fitness only slightly: 0.323 and 0.328, respectively, compared with 0.329 for the complete method. Fixed spatial-greedy reaches 0.348, whereas the other fixed policies are 0.02–0.08 lower than the complete method. Thus, the proxy does not show an advantage for switching over the strongest fixed policy. The matched Gazebo ablation tests that policy separately.

A second proxy ablation isolates geodesic travel-time estimation. Across 500 seeds, replacing Euclidean distance with geodesic distance in the allocation cost improves fitness by 2.13, 1.83, and 4.10 percentage points in the three scenarios ($p < 10^{-4}$). Effect sizes are negligible to small ($\delta = 0.128$–0.275). Geodesic arrival-time estimation therefore provides a consistent but modest improvement in the proxy. It is not isolated in Gazebo.

The matched closed-loop ablation applies the planned and exploratory configurations from Section V-C to 20 common seeds at 5r/25t.

Under deadline pressure, the complete method scores 0.650, compared with 0.556 for fixed EDF and 0.540 for fixed orphan-first. Both reductions are large and remain significant after correction ($p < 0.005$). Removing the override cascade yields 0.641, with no detectable difference from the complete method ($p_{\text{adj}} = 1.000$, $\delta = 0.06$).

The exploratory fixed spatial-greedy configuration also does not differ detectably from the complete method (0.638 versus 0.650, paired $p = 0.43$, $\delta = 0.09$). This non-significant comparison is not an equivalence test. The results show that fixed EDF and orphan-first perform poorly in this benchmark. They do not establish an advantage for the implemented selector over the strongest fixed heuristic. Such a claim requires evaluation in regimes where spatial-greedy is not competitive.

VII. DISCUSSION

A. Interpretation and Practical Implications

The three evaluation settings address distinct sources of system behavior. Allocation-only simulation tests the event-level assignment rule. The stochastic proxy adds uncertain goal execution, and Gazebo adds congestion, local planning, and recovery. Changes in method ranking across these settings support the need for closed-loop validation. They do not identify any single navigation mechanism as the cause.

The proxy objective reduces the distinction between late and unfinished tasks. In contrast, the closed-loop allocator may continue overdue tasks to improve completion. Proxy fitness and closed-loop performance can therefore produce different margins. The supplementary artifact provides detailed payload, trajectory, efficiency, and timeline analyses.

B. Limitations

Corrected inference is strongest at the primary fleet size. Results from the pooled 3-robot campaign and the 10-robot cells are descriptive. The high-seed proxy experiments support only proxy-level claims. Broader inference requires additional closed-loop seeds at both secondary fleet sizes.

TABLE VII

MATCHED CLOSED-LOOP ABLATION AT THE PRIMARY 5-ROBOT / 25-TASK SCALE: FOUR ARMS OVER 20 COMMON SEEDS PER SCENARIO, PRE-REGISTERED BEFORE THE CAMPAIGN (PAPER/PREREGISTRATION.MD). PRIMARY ENDPOINT IS EFFECTIVE ON-TIME COMPLETION $\mathrm{CR} \times (1 - \mathrm{DVR})$ UNDER *deadline_pressure*; p IS A PAIRED WILCOXON SIGNED-RANK TEST AGAINST *full*, BONFERRONI-CORRECTED WITHIN THE SCENARIO FAMILY, AND δ IS CLIFF'S DELTA WITH A BOOTSTRAP 95% CI. THE OTHER TWO SCENARIOS ARE DESCRIPTIVE.

	full	**no-override**	**fixed-EDF**	**fixed-orphan**
deadline_pressure – primary endpoint				
effective on-time	0.650	0.641	0.556	0.540
p (Bonferroni)	–	1.000	<0.005	<0.005
Cliff's δ vs full	–	+0.06	+0.57	+0.62
robot_failure – descriptive				
effective on-time	0.986	0.998	0.846	0.850
completion rate	1.000	1.000	0.998	1.000
recovery time (s)	83.7	63.3	88.5	56.6
mixed_stress – descriptive				
effective on-time	0.616	0.606	0.590	0.601

External validity is limited by one arena, homogeneous robots, and ground-truth-anchored localization. Host resources also prevented validation beyond the 10-robot fleet. Thus, the largest tested fleet reflects the available ROS 2/Nav2 resources rather than a demonstrated algorithmic scaling limit.

The benchmark compares configured pipelines rather than equally tuned abstract algorithms. AHE-MRTA was developed for the reported arena. The baselines were implemented from their source descriptions without comparable tuning. In addition, some seeds used for proxy tuning overlap with the reported campaign.

Component attribution remains incomplete. The dominance fallback selects one policy in almost every logged state that reaches it. Geodesic travel estimation is isolated only in the proxy, and the designer-specified mapping matrices are not learned. The ablations therefore evaluate specific alternatives but do not prove that the complete selector is necessary.

The proxy intentionally uses a harsher navigation-failure model than Gazebo, so it measures relative allocation robustness rather than physical success. The implemented priority rescaling can also reverse the intended ordering. The Supplementary Material documents this issue and other implementation diagnostics.

VIII. CONCLUSION

This paper presented an event-triggered MRTA pipeline that combines a compact policy portfolio, geodesic arrival estimates, and bounded load repair. In both simulation settings and at the primary closed-loop scale, the complete pipeline achieved the highest observed completion and deadline performance while remaining within the decision-time target.

Matched ablations narrow the causal claim. Access to a suitable policy is important. However, neither the override cascade nor the complete selector showed a detectable difference from the dominant spatial-greedy fallback in the evaluated regime. The evidence therefore supports access to a suitable policy within the portfolio, but not the superiority of the implemented switching rule.

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